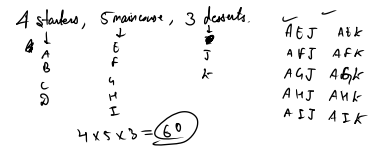


Counting
↳ How many ways can something happen?

Fundamental counting principle → multiplication rule

task 1 → m ways
task 2 → n ways
↳ m x n ways



Coin → H or T → 3 times → 2 x 2 x 2 = 8

HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

ATMs → 4 digits pin

X X X X
↓ ↓ ↓ ↓
0-9 0-9 0-9 0-9
10 x 10 x 10 x 10

10,000 → possible pins

0
1
2
3
4
5
6
7
8
9

lowcase

→ only letters
length = 8

total lowcase letters = 26⁸ ⇒ billions

odd uppercase

26 + 26 = 52 ⇒

↳ odd digits + symbols. ASCII ⇒ 95⁸ → quadrillions

length matters more than complexity

→ brute force attack. → hacker tries every possible combination
bigger search space → harder attack

Permutations

→ order matters.

n!

$n P_r = \frac{n!}{(n-r)!}$

15 = 5 x 4 x 3 x 2 x 1 = 120

5 people

P1 P2 P3 P4 P5

5 choices
4 choices
3 choices
2 choices
1 choice
⇒ 120

recommendation system

move A → move B → move C
move C → move A → move B

10 students
team of 3

A, B, C
C, B, A
B, C, A

combination → order not matter

$n C_r = \frac{n!}{r!(n-r)!}$

Suppose dataset → 20 features

best 5 features → 20 C₅

⇒ $\frac{20!}{15!5!} = \frac{20 \times 19 \times 18 \times 17 \times 16}{1 \times 2 \times 3 \times 4 \times 5} = 15504$

ML → greedy methods

Pascal Triangle

binomial coefficients

1, 4, 6, 4, 1
1 C₀, 4 C₁, 6 C₂, 4 C₃, 1 C₄
1 4 6 4 1

1 4 6 4 1
C(4,0) -
C(4,1) -
C(4,2) -
C(4,3) -
C(4,4) -

P → 2 heads in 4 tosses

total p = 2⁴ 4 C₂ = 6

Coin → 4 times

↳ How many times exactly 2 heads?

row 4 → 1, 4, 6, 4, 1

→ 0 1 2 3 4

→ C → H, T

let → H

HHHH
HHHT
HHTT
HTTT
TTTT
THHH

(H)
(T)

6 → HHTT
HTHT
HTTH
THTT
THTH
TTTH

1 → 0 heads possible in 1 way
4 → 1 heads possible in 4 ways
6 → 2 heads possible in 6 ways
4 → 3 heads possible in 4 ways
1 → 4 heads possible in 1 way

Probability

↳ language of uncertainty

Uncertain

Deterministic system

Same input → same output
2 + 5 = 7

Stochastic system

Same input → diff output every time
PROBABILITY

ML Algorithm

Probability means

Real world

Naive Bayes classifier

P(spam | words in mail)

Spam filter

Logistic Regression

P(click | user features)

Google ads click

Neural network (deep learning)

P(class | image pixels)

Image object recognition

Hidden Markov model

P(word sequence)

Voice assistants (Alexa, Siri)

Sample Space

ω

↳ All possible outcomes (Ω)

Ω = {1, 2, 3, 4, 5, 6}

Ω = {H, T}

Ω = {spam, not-spam}

Event

⇒ Ω = {1, 2, 3, 4, 5, 6}

A → even numbers = {2, 4, 6}

sum of S.S.

⇒ $\frac{3}{6} = \frac{1}{2} = 0.5$

Probability range

0 ≤ P(A) ≤ 1

→ 0 means impossible

→ 1 means certain

→ between 0 & 1 means uncertainty

KOLMOGOROV AXIOMS

probability 3 fundamental rules

(1, 2, 3, 4, 5, 6)

$\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$

1. Non-negativity

2. Total probability equals to 1

P(S) = 1

3. Additivity axiom

P(A ∪ B) = P(A) + P(B)

if A ∩ B = ∅

→ A & B are mutually exclusive

Class
Cats 0.7
Dogs 0.2
Horses 0.1
X = 1
0.7 + 0.2 + 0.1 = 1

A → 2 → $\frac{1}{6}$

B → 5 → $\frac{1}{6}$

2 or 5 = $\frac{1}{3}$

P(A ∪ B) = P(A) + P(B)

J.T.

Genuine

Fraud

Prob.

0.95

0.05

1 → Yes

2 → 0.95 + 0.05 = 1

3 → G & F → mutually exclusive

Bayesian ML

$P\left(\frac{A}{B}\right) = \frac{P(B|A)P(A)}{P(B)}$

Start with an initial belief → observe data → update belief

before test → P(Disease) = 0.02 → prior belief

P(Disease) = 0.98

Classical Probability

$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$

Frequentist probability

India win 63 out of 100 matches

P(India win) = 0.63

Bayesian probability